

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging
10 marks

Q2

Hospital Registration –
NullPointerException
Debugging
10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging
10 marks

Q4

Bank Account – Race Condition
Debugging
10 marks

Q5

Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q2

Hospital Registration –
NullPointerException Debugging10
marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter patient name: ");
        String inputName = sc.nextLine();
```

```
String name;

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
}
```

The corrected program must prevent
NullPointerException and handle invalid age input
without terminating unexpectedly.

✳ **Debugging Task:** Find and fix the bugs in the code
shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2
3 class Main{
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         String name = null;
8         int age = 0;
9
10        try{
```

```
11         System.out.print("Enter patient
12         String inputName = sc.nextLine()
13
14         if(inputName == null || inputName
15             name = null;
16             throw new NullPointerException
17         }else{
18             name = inputName;
19         }
20
21         System.out.print("Enter patient
22         if(sc.hasNextInt()){
23             age = sc.nextInt();
24             if(age<0 || age>120){
25                 throw new IllegalArgument
26             }
27         }else{
28             throw new IllegalArgumentExceptionExc
29         }
30
31         System.out.println("Registration
32     }catch(NullPointerException e){
33         System.out.println("Error: "+e.g
34     }catch(IllegalArgumentExceptionException e){
35         System.out.println("Error: "+e.g
36     }catch(Exception e){
37         System.out.println("Runtime erro
38     }finally{
39         System.out.println("Registration
40         sc.close();
41     }
42 }
43 }
```



Test Input

Bob miller
-5

Output

```
Enter patient name: Enter patient age: Error:  
Invalid age range.  
Registration completed
```

 Pending manual review

 Submitted

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